

Question:

John drives for 3 hours at a speed of 60 mph and then turns around because he realizes he forgot something very important at home. He tries to get home in 4 hours but spends the first 2 hours in standstill traffic. He spends the next half-hour driving at a speed of 30mph, before being able to drive the remaining time of the 4 hours going at 80 mph. How far is he from home at the end of those 4 hours?

Reasoning + Answer:

<think john first travels for 3 hours at 60 mph so the distance covered is $3 * 60 = 180$ miles , now he turns back so the distance from home is also 180 , so now he has 4 hours to reach home , but he is in stand still traffic for 2 hours so the effective time he has to cover the distance is $4 - 2 = 2$ hours , he covers the first 0 . 5 hours at 30 mph so the distance covered in that time is $0 . 5 * 30 = 15$, so the remaining distance to cover in 1 . 5 hours is $180 - 15 = 165$, at a speed of 80 , so the distance covered in 1 . 5 hours = $80 * 1 . 5 = 120$, so the distance left at the end is $165 - 120 = 45$ </think><answer>45 </answer>>

